

1 / What I propose for the pins

Start with 132 FPGA signal pins for the ASICs. This assumes eight recording chips, four of which also stimulate. Keep the recording inputs and local stimulation controls separate. Share only the controls below that the team agrees can operate together.

Keep these 88 inputs separate

Signal	Pins	Why it stays
Data<1:8>	64 in	Eight data wires per chip. Joining outputs from different chips is not a way to save pins.
CLK32MHz_Out	8 in	One returned clock per chip, so the FPGA can capture that chip's data at the right time. Use suitable clock-input pins.
Read, Sync	16 in	Two markers per chip, so the FPGA can identify the data sequence and channel positions.

These functions are documented in the ASIC slides. Keep them for the present interface; Gerald must confirm both ASIC versions.^[1]

Controls: the choice, the benefit, and the concern

Signal / pins	Proposed choice and benefit	Concern and what we need to do
CLK32MHz_In 1 out	Use one shared recording clock. We save seven pins and give all eight chips the same clock rate.	The clock must reach all chips reliably. Zitong checks the distribution circuit and wiring. Use eight outputs only if separate clock control or routing requires them.
Chip_Reset 1 out	Start with one shared chip reset. We save seven pins and can restart all chips together.	Resetting one chip would also reset the other seven. Ask whether that recording interruption is acceptable. If not, use eight resets: +7 pins .
FE_RESET 8 out	Keep one amplifier reset per chip. Resetting one chip's amplifiers will not force a reset of all the others.	This costs seven more pins than sharing. Keep them: the legacy stimulation sequence uses amplifier reset, so sharing could blank every chip. ^[2]
SPI CLK, LATCH 2 out	Start with shared CLK and LATCH. This reduces wiring. LATCH is the signal that tells the chips to apply their waiting settings.	A shared LATCH applies settings on every chip. Jiaao must have valid settings ready for all of them. Otherwise, consider eight latches: +7 pins , after Gerald confirms that this works.
SPI L, R 16 out	Keep two programming wires per chip. We can give different chips different gain and stimulation settings.	These use 16 pins. Do not share them unless Gerald confirms another way to program chips individually.
STIM_EN, STIM_START, STIM_CHB 12 out	Keep enable, trigger and charge-balancing control separate for each of the four stimulating chips. We can control one without sending the same commands to all four.	This uses 12 pins. Gerald must specify the correct sequence and startup levels. Keep stimulation disabled during boot; use the ASIC-approved charge-balancing state.
STIM_CLK 4 out	Keep one clock output per stimulating chip. We can control each chip's clock separately.	This costs only three more pins than sharing. Keep four for the first board; share only if the team agrees all four can use the same clock behavior.

Total: 88 inputs + 44 outputs = 132 ASIC-facing signal pins. If both chip reset and LATCH become independent, the total becomes **146**. This does not include the controller link, boot flash, reference clock, or power/ground contacts.

Sources: [1] ASIC slides 13–17 and 25–28; [2] legacy stimulation code. Sharing choices are proposals. Separate amplifier resets do not guarantee that other chips' recordings are free of stimulation artifacts.

2 / Which FPGA should we choose?

Evaluate XC7A35T-2CSG324C first. It is the practical starting candidate from the checked stock. Choose XC7A50T-2CSG324C instead if a supplier confirms delivery and the added capacity is worth the price. **Do not commit the bulk order until the design and controller link pass the checks below.**

Exact part	If we choose it, we gain. . .	But the concern is. . .
XC7A35T-2CSG324C First candidate	A 15 × 15 mm package with 210 signal I/O. The preliminary voltage-group plan can accommodate 132 ASIC signals without removing useful controls. ^{[3][6]}	We have not yet proved the full eight-chip design fits or runs fast enough. It also lacks dedicated GTP high-speed link hardware. ^[3]
XC7A50T-2CSG324C Upgrade if available	More logic and internal memory, with the same package size and 210 I/O. This gives more room for the FPGA design. ^[3]	Delivery is not confirmed. It also has no GTP, so upgrading from 35T to 50T does not solve a GTP-link requirement. ^[3]

Stock check: what the supplier actually shows

DigiKey US, checked 23 September 2026: 35T lists **249 in stock at \$70.46 each**; 50T lists **0 in stock at \$88.50 each**, with 126 expected on 23 August 2027. These are displayed one-piece prices and inventory, not a confirmed delivery quote.^{[4][5]}

Action: ask for written quotes for **100 plus spares** of both exact parts. An authorized supplier may have another source of 50T stock, but do not plan around it without confirmation. Do not substitute CSG325 or a different speed grade without checking the design again.

What we need to choose and check to make it work

Suggested reviewer	Question to close / required result
Gerald + Jiaao	Can chip reset and LATCH be shared? Confirm both ASIC versions, whether restarting all eight chips is acceptable, and whether settings can be applied to every chip together. That fixes 132 versus 139 or 146 pins.
David + Jiaao	How will all the data reach the controller? Choose the receiving interface, cable and data format. With 16 transmitted bits per sample, the proposed 4K system produces 2.048 Gbit/s before extra packet data . A connector name alone does not prove it can carry that stream.
Jiaao	Does the complete design fit the 35T and meet timing? Test all eight chip interfaces, stimulation, the controller link and buffers together in Vivado. Decide how much buffering is needed when the receiver pauses. Do not remove extra memory before checking this.
Howard + Zitong	Can we wire it correctly and fit it on the board? Assign the signals to pins with compatible voltages and clock functions. Include boot flash, clock source, controller connections, power supplies and startup behavior. Confirm connector size, board size and expected power/heat.

One decision must come before buying: does the controller link require GTP?

If it does, **neither of these CSG324 parts is suitable unchanged**. Choose a different package/link architecture. If it does not, the 35T remains the first candidate, subject to the full-design checks above.^[3]

Sources: [3] AMD DS180, Artix-7 tables; [4–5] linked DigiKey part pages; [6] AMD 35T-CSG324 package map. Signal functions: repository sources [1–2] on page 1. The 100-unit target and 4K system plan come from the supplied September 23 meeting notes. No new logic-fit, timing, electrical or hardware validation was performed for this wording revision.